

F1AUDIO: A Formula-1 Engine Audio Dataset

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Abstract—We introduce F1Audio, a novel dataset of features extracted from Formula-1 onboard engine audio, synchronized with car telemetry comprising approximately 300 hours of data from 10 Grand Prix events of the 2024 Formula-1 season. This is the first large-scale dataset combining engine audio features with vehicle speed, engine revolutions per minute (RPM), gear selection, and throttle percentage. We also provide benchmark results for the prediction of telemetry targets from the engine audio.

Index Terms—Formula-1, engine audio, telemetry estimation, dataset, audio features

1. INTRODUCTION

The engine audio of a Formula-1 car is a rich acoustic signal. Engine RPM, throttle position, and gear selection are all reflected in the spectral content and time-varying envelope of the exhaust note (the characteristic sound produced by the engine’s exhaust system), and experienced engineers can identify gear changes or throttle blips (brief, deliberate increases in engine RPM during downshifts) by ear [1].

In the modern Formula-1 ecosystem, both high-frequency telemetry and onboard audio are publicly available through official APIs and broadcast feeds, respectively. However, they are disseminated through different channels, resulting in data that are not temporally aligned and are therefore difficult to combine. This creates a barrier to research on motorsport acoustics. With few existing automatic approaches to extract information from engine sound, large-scale open datasets are needed to facilitate more work in the area. To date, no public dataset provides synchronized, high-resolution audio-derived features and telemetry from Formula-1 cars.

The consumption of modern motorsport media is increasingly driven by a desire for granular data and immersive experiences. Observations of driver point-of-view footage reveal that the visibility of steering wheel displays provides immediate context regarding vehicle performance, significantly enhancing the viewer’s engagement.¹ Acoustic telemetry recovery enables applications such as enriching archival race footage with estimated telemetry overlays, supporting motorsport analysis where telemetry is unavailable, and informing procedural audio generation for simulation and game development.

The primary contribution of this work is F1Audio, a large-scale dataset of features extracted from Formula-1 onboard audio, synchronized with telemetry from the 2024 season. In addition, we

provide baseline models and results for estimating speed, RPM, gear, and throttle from audio embeddings and analyze the effect of temporal context.

2. RELATED WORK

The analysis of automotive and motor acoustics spans several research areas, including acoustic fault diagnosis, vehicle speed estimation, and acoustic scene classification. This section reviews prior work most relevant to our goal of recovering vehicle telemetry from engine audio.

2.1. Acoustic RPM and speed estimation

Classical signal processing techniques have long been applied to extract kinematic parameters from mechanical and acoustic engine emissions. Engine RPM estimation via harmonic tracking—following the engine’s fundamental frequency and its harmonics—has been explored since the early 1990s. Early foundational work relied on physical vibration sensors, utilizing methods such as the Vold-Kalman order tracking filter to perfectly isolate rapidly changing mechanical frequencies [2]. Over the following decades, these techniques were successfully adapted to purely acoustic settings, establishing that the same kinematic properties tracked by physical sensors are fully preserved in the audio signal. Modern examples of this transition include estimating RPM directly from microphone signals using Kalman-based frequency smoothing [3]. In other related work, vehicle speed estimation in “drive-by” scenarios has leveraged acoustic Doppler shift analysis [4].

2.2. Acoustic condition monitoring and vehicle diagnostics

Beyond RPM and speed estimation, audio has been used for engine identification—for example, differentiating petrol from diesel engines using time-domain features [5] or cascading neural networks for fuel type, engine configuration, and cylinder count [6]. Acoustic fault detection is also well established: machine learning classifiers trained on spectrograms can identify mechanical failures such as misfiring cylinders or bearing faults [7], and recent work has demonstrated the value of psychoacoustic metrics for diagnosing gear transmission noise [8].

Similar principles apply to electric motors, where the research focus has increasingly shifted toward identifying robust acoustic feature representations for condition monitoring. To capture these mechanical states, representations such as Mel-frequency cepstral coefficients (MFCC), Short-Time Fourier Transform (STFT), and Mel spectrograms have been investigated on simulated motor data [9]. More specialized representations such as wavelet-based features have been reported to offer advantages by localizing transient events in the time-frequency domain, which could be beneficial for detecting intermittent faults [10]. Subsequently, modern approaches couple these feature representations with deep learning architectures, such as combining self-organizing maps with long short-term memory networks to diagnose rotary motor faults under varying loads [11].

¹For discussions on fan engagement through telemetry and AI, see online reports from SportsPro, “How is AI changing the way fans watch Formula One?” 2024. [Online]. Available: <https://www.sportspro.com/analysis/sponsorship-marketing/f1-broadcasts-aws-artificial-intelligence-generative-ai/> (Accessed: 11 June 2026); IBM, “IBM + Scuderia Ferrari perfect a new formula for fan engagement,” 2025. [Online]. Available: <https://www.ibm.com/think/insights/ferrari> (Accessed: 11 June 2026); SVG Europe, “Motorsport innovator Griip grasps racing by the data to engage fans with cloud-based technology,” 2022. [Online]. Available: <https://www.svg-europe.org/blog/headlines/motorsport-innovator-griip-grasps-racing-by-the-data-to-engage-fans-with-cloud-based-technology/> (Accessed: 11 June 2026); and Autosport, “How is artificial intelligence changing Formula 1?” 2024. [Online]. Available: <https://www.autosport.com/f1/news/how-is-artificial-intelligence-changing-formula-1/10659532/> (Accessed: 11 June 2026).

The methodological overlap between automotive diagnostics and the broader field of industrial sound analysis provides a robust foundation for this work. Studies in industrial acoustic monitoring demonstrate that machine learning techniques, when applied to complex machinery noise, are highly effective at identifying subtle operational states [12]. Motivated by previous work, we extend these acoustic analysis tasks to high-performance Formula-1 engines and the task of continuous recovery of multi-faceted telemetry data. While acoustic monitoring and datasets are well-established for basic industrial fault detection and simple engine RPM tracking, other telemetry data is rarely worked on

2.3. Related datasets

Several public datasets have some relation to research in automotive and motor acoustics. The ESC-50 dataset [13] provides environmental sound recordings, including vehicle sounds, but lacks telemetry. The DCASE challenge series offers tasks on acoustic scene classification and sound event detection, but has yet to focus on vehicles. For industrial motor monitoring, datasets of miniature-machine operating sounds (such as ToyADMOS) [?] Thus, there is a clear lack of publicly available data pairing automotive engine derived features with telemetry.

3. THE F1AUDIO DATASET

3.1. Data acquisition and sources

3.1.1. Onboard engine audio data: Onboard engine audio was obtained from F1 TV, the official Formula-1 broadcast subscription service. The audio signal was captured from the broadcast stream using the OBS Studio application, which records the system audio directly from the broadcast feed during the race. The onboard audio transmitted by F1 TV is provided as a single-channel, mono source, which contains engine sound, mechanical events, and team radio communications. Manual spot-checks confirm that this broadcast mix is uniform, with no noticeable variation in levels across cars from the same team as well as across different drivers, teams, or circuits. Each driver’s onboard channel was recorded, yielding one mono audio file per driver per race at 48 kHz and 16-bit word length. F1 TV’s terms of service permit personal, non-commercial use of content for educational and research purposes under the fan usage provisions of the official Formula-1 intellectual property guidelines;² all source audio used to generate the F1Audio dataset was processed solely for academic research.

3.1.2. Telemetry Data: Car telemetry was retrieved using FastF1,³ a publicly available Python library interfacing with the Formula-1 broadcast service.

The data returned by FastF1 constitutes the **raw telemetry data** released in the dataset: it is provided as-is, with *non-equidistant* timestamps reflecting the variable polling intervals of the broadcast telemetry sensors present on each car. Telemetry is returned in CSV format with the following channels included in F1Audio:

- **Time:** race clock in seconds, initialized to 0.0 at race start.
- **Lap:** current lap number.
- **RPM:** engine revolutions per minute (range: $\approx 3\,000$ – $13\,000$).
- **Speed:** vehicle speed in km h^{-1} (range: 0–350).
- **nGear:** selected gear (0 = neutral, 1–8 = drive gears).
- **Throttle:** throttle input as a percentage (0 - 100).

²Formula 1, “Intellectual Property Guidelines.” [Online]. Available: <https://www.formula1.com/en/information/guidelines.4EOKE9RRqevL4niTK9kWyT> (Accessed: 11 June 2026).

³P. Schaefer, “FastF1: A Python Package for Formula 1 Data.” [Online]. Available: <https://docs.fastf1.dev/> (Accessed: 11 June 2026).

Table 1: Example of raw telemetry data retrieved via FastF1.

Time	Lap	RPM	Speed	nGear	Throttle	Brake
56.000	1.0	10681.9	258.84	6	100.00	FALSE
56.419	1.0	10967.0	262.34	6	100.00	FALSE
56.930	1.0	9445.7	251.54	5	89.78	TRUE
57.290	1.0	8987.4	202.02	5	34.64	TRUE
57.638	1.0	6574.0	147.97	5	0.00	TRUE

- **Brake:** braking status (TRUE or FALSE).

The sample rate of the telemetry data varies between approximately 4 Hz and 8 Hz depending on sensor polling intervals and broadcast network conditions. Table 1 provides an example of this unprocessed raw telemetry, demonstrating the irregular polling intervals.

3.2. Data processing

The raw data—OBS-captured audio files and acquired FastF1 CSV telemetry—undergo a series of processing steps to ensure consistency and synchronization. The processing pipeline consists of temporal alignment, normalization, cleaning, telemetry interpolation, and feature extraction. Because licensing restrictions prohibit the distribution of raw broadcast audio, the dataset is released as a collection of pre-extracted audio features and aligned telemetry. Consequently, the processing steps are strictly partitioned. The **Data Pipeline** (temporal alignment, audio preprocessing, cleaning, and feature extraction) describes how the released dataset was generated from the raw proprietary audio. The **Experiment Pipeline** (Section 4) describes how these released features are subsequently modeled.

3.2.1. Temporal alignment: Audio recordings are manually trimmed to the race start (lights out), establishing a common time reference with the telemetry clock.

3.2.2. Audio preprocessing (Data Pipeline): The audio undergoes standardized preprocessing before feature extraction to ensure the resulting feature dataset is consistent. Each recording is resampled to 32 kHz to meet the input requirements of the PaSST architecture. Before feature extraction, we normalize the audio Root Mean Square (RMS), computing a global gain factor per race recording to ensure consistent loudness. We set the target RMS level to 0.1 to provide approx. 20 dB of headroom for engine transients. While distributing unprocessed data is generally preferable to avoid artifacts, the necessity of releasing a feature-only dataset requires this upfront normalization. This strategy leads to individual short sections of audio exceeding the clipping threshold. However, only approximately 0.02% of samples exceed this threshold, and we found that the clipping almost exclusively occurs in brief radio communication bursts.

3.2.3. Data cleaning: Incomplete sessions (e.g., cars crashing or retiring on the opening lap) are entirely filtered out. For late retirements, data is retained up to the point of mechanical failure to ensure only valid engine acoustic data is processed.

3.2.4. Telemetry alignment and label generation: Because the raw telemetry is sampled at an irregular rate of approx. 4 Hz to 8 Hz, it is interpolated onto a uniform time grid with a target temporal resolution of 50 ms. Continuous telemetry values are obtained by linear interpolation of the CSV data over this new high-resolution grid. Discrete variables, such as gear and braking status, are aligned using nearest-neighbor assignment. If the raw telemetry stream begins slightly after or ends slightly before the audio timeline, the first and last valid values are repeated to pad these unmeasured extreme boundaries. This fully aligned, high-resolution telemetry is provided as an additional manifest file within the dataset, complementing the raw sensor logs.

Table 2: Global statistics for the F1Audio dataset.

Property	Value
Total Audio Length in Dataset	266h 11m 30s
Total Feature Frames	1,914,020
Number of Audio Files	197
Number of Race Tracks	10
Number of Teams	10
Train/Test Dataset Audio Length	210h 17m 31s / 55h 53m 59s
Feature Frames in Train/Test Dataset	1,511,598 / 402,422

3.2.5. PaSST feature extraction: We use the Patchout Fast Spectrogram Transformer (PaSST) [14], a Vision Transformer pre-trained on AudioSet, as a fixed feature extractor. The architecture produces a 1295-dimensional embedding every 50 ms, yielding 20 embeddings per second. Because PaSST’s positional encoding is limited, each 90 min race file is split into consecutive non-overlapping 9 s chunks, and the resulting embeddings are concatenated to reconstruct the full timeline. The dense resolution allows downstream users to construct arbitrary analysis windows without reprocessing the audio.

3.3. Dataset structure

The dataset employs a standardized naming convention where the raw telemetry files are named using the format `Team_Driver_RaceTrack`, effectively encoding session metadata directly into the file system.

The `telemetry_raw/` directory contains the original telemetry logs as acquired from the FastF1 API. For each Grand Prix event, aligned features and labels are serialized into three distinct output files:

- **Features (.h5):** Contains PaSST embeddings.
- **Manifest (.parquet):** A unified tabular file where raw telemetry (RPM, Speed, Gear) has been resampled and aligned to the exact 50 ms grid of the audio features.
- **Summary (.json):** Provides per Grand-Prix folder metadata, including total audio duration and class distributions for drivers and teams.

3.4. Dataset statistics

The overall dataset scale and properties are summarized in Table 2. Two race tracks (Miami and Bahrain) are held out entirely as an unseen test set on which the benchmark results are reported. The remaining eight tracks are used for training and validation. Within these eight tracks, the individual feature frames are randomly partitioned into training (80%) and validation (20%) sets. To ensure strict reproducibility and framework-agnostic usage, the exact frame-level assignments are explicitly defined in a static metadata manifest distributed alongside the dataset.

Across the 10 Grand Prix events, the recording length per track ranges from roughly 21 to 32 hours, with a median duration of approximately 28 hours. The dataset also provides a relatively balanced distribution across the 10 constructor teams, with cumulative audio lengths ranging from roughly 21 to 28.5 hours per team, and a median of 24 hours.

Figure 1 visualizes the statistical distributions of core telemetry variables across the entire dataset. Key observations from these telemetry distributions include: bimodal throttle values with high mass at 0% and 100% (reflecting the all-or-nothing nature of racing throttle application); non-uniform gear usage owing to the track layouts of the selected race tracks; and a brake-to-non-brake ratio of approximately 1:4, indicating significant class imbalance; The plot reveals a predominance of high RPM values, as these cars are engineered to output peak power in that upper band.

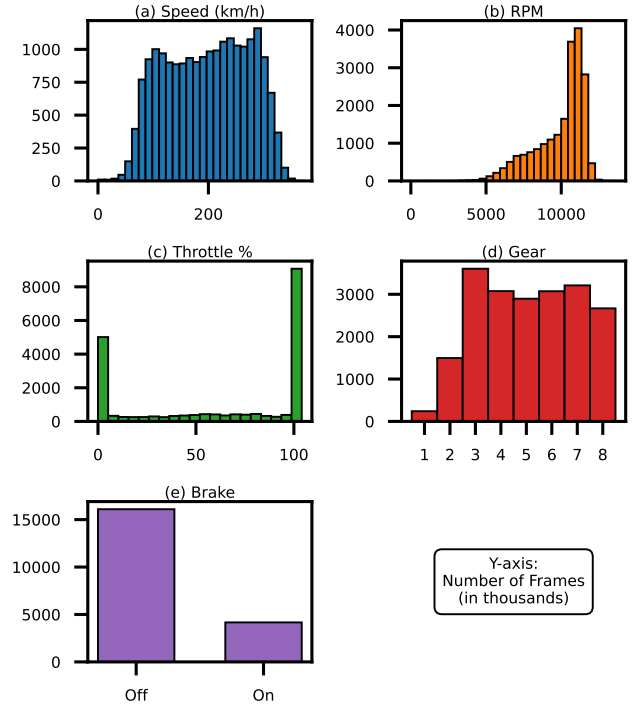


Fig. 1: Telemetry distributions in the F1Audio dataset: broad operating ranges for speed (a) and RPM (b), throttle (c) bimodal, gear usage (d) skewed towards high gears, and imbalanced brake classes (e).

3.5. Data availability and license

The dataset, including all telemetry files, pre-extracted PaSST features, and manifests, is released publicly under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International (CC BY-NC-ND 4.0) license.⁴ The data is intended solely for academic use, in accordance with F1 TV’s terms of service. Due to licensing restrictions and the non-commercial terms of use of the source broadcasts, *F1Audio* is released as a feature-only dataset.

4. EXPERIMENTS AND RESULTS

4.1. Experimental setup

All models in this section are trained on the pre-extracted PaSST feature windows described in Section 3.2.5. The training/validation and test splits are fixed across all experiments. To gain core insights on the data and its information content, and to establish basic benchmark expectations, we provide results for two classifiers and one ablation study:

- **Baseline Models (LP):** Establishing performance lower bounds for telemetry recovery and evaluating team-specific acoustic signatures using linear probes.
- **Multi-Layer Perceptron (MLP):** Evaluating independent feed-forward models on isolated 1 s analysis windows to establish task-wise telemetry prediction performance.
- **Ablation Study: Temporal Context:** Assessing how the inclusion of neighboring feature frames (variable-length context windows) affects telemetry regression accuracy.

⁴Dataset is available at: <https://doi.org/10.5281/zenodo.21186137>

Table 3: Telemetry prediction and team classification benchmark results on held-out test tracks.

Model	RPM (R^2)	Speed (R^2)	Throttle (R^2)	Gear (Macro F1)	Team (Macro F1)
LP (1 s)	0.60	0.71	0.40	0.34	0.83
MLP (1 s)	0.68	0.76	0.52	0.39	0.92
MLP (5 s)	0.78	0.85	0.69	0.52	0.96
MLP (9 s)	0.79	0.86	0.69	0.54	0.95

4.1.1. Benchmark Preprocessing and Metrics: To establish baseline performance and evaluate whether engine audio encodes vehicle state, we predict three continuous variables (RPM, speed, and throttle) and two discrete variables (gear and constructor team). For the input representation, PaSST embeddings are grouped into 1 s analysis windows with a 0.5 s hop size by averaging 20 consecutive 50 ms embeddings per window. Prior to training, all model inputs and continuous targets are standardized to zero mean and unit variance using a scaler fitted solely on the training data.

To ensure comparability, both the linear baselines and MLP models optimize identical loss functions appropriate for each target. Regression tasks (RPM, speed, and throttle) optimize the Mean Squared Error (MSE). In case of throttle, the MSE is weighted inversely to the frequencies of 20 equal-width bins to counteract its strongly bimodal distribution. Classification tasks (gear and team) optimize cross-entropy loss. Gear classification uses an unweighted loss, while team classification applies balanced class weights (inverse frequency) to compensate for unequal samples per team.

Model performance is evaluated using the coefficient of determination (R^2), while both discrete classification tasks are reported using the macro-averaged F1 score.

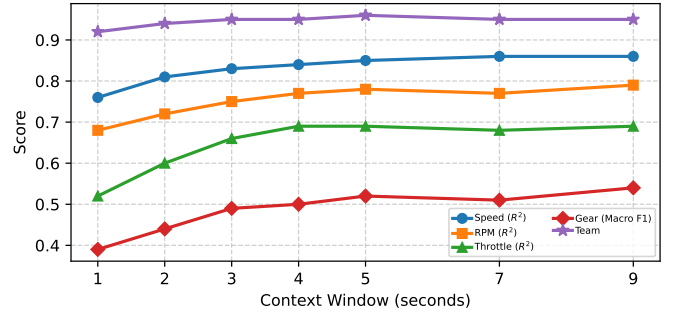
4.1.2. Baseline Models: Two simple linear baselines gauge the difficulty of the telemetry recovery tasks and establish performance lower bounds. For each continuous regression target (RPM, speed, throttle), we train a separate ridge regression model. For the discrete classification tasks (gear and team), we employ multinomial logistic regression. Both baselines operate on identically standardized 1 s feature windows.

4.1.3. Multi-Layer Perceptron (MLP) Models: Five separate feed-forward MLP models are trained—one for each of the core telemetry stems and the constructor team. Each shares a common trunk of two fully connected layers (256 and 128 units) with ReLU activations and dropout ($p = 0.2$), feeding into either a single-node regression head or a K -class classification head. To ensure a fair, direct comparison with the linear probes, the default input vector for all five models represents an isolated 1 s feature window (an architectural choice systematically ablated in Section 4.2.1). All models are trained using the Adam optimizer ($lr = 10^{-3}$) and early stopping monitoring validation loss. Full implementation details are available in the accompanying repository.⁵

4.2. Results

Table 3 reports the performance of both the linear baselines and the MLP models on the unseen held-out test circuits for telemetry prediction. Additionally, the baseline linear probe was evaluated on the discrete task of team classification, achieving a robust Macro F1 score of 0.83.

4.2.1. Ablation study: Effect of temporal context: We investigate the impact of the input context length to determine if providing the model with a short acoustic trajectory improves prediction

Fig. 2: Effect of context window length on MLP performance (R^2 for continuous targets, Macro F1 for Gear & Team).

over instantaneous frames. For a given context of C seconds, we concatenate neighboring PaSST embeddings to form the flattened input vector, padding the temporal boundaries with zeros where necessary. We evaluate context windows of $C \in \{1, 2, 3, 4, 5, 7, 9\}$ seconds for the MLP configuration. As visualized in Figure 2, increasing the temporal context steadily improves performance across all targets before plateauing between 5 s and 9 s.

4.3. Discussion

The results demonstrate that acoustic telemetry recovery is feasible. While the linear probe establishes a solid baseline, introducing the non-linear transformations via the MLP yields immediate performance gains across all continuous stems, notably improving throttle prediction.

Still, several tasks remain challenging. Throttle prediction struggles because throttle changes do not instantaneously alter acoustic pitch due to mechanical inertia. Similarly, gear classification yields a low Macro F1 due to kinematic ambiguity and temporal blurring: unlike speed or RPM, multiple gears share identically overlapping RPM ranges at vastly different vehicle speeds. Because engine acoustics are largely determined by RPM, the model struggles to disambiguate the selected gear from an isolated audio frame.

The ablation study confirms that temporal context partly addresses this challenge by providing a continuous kinematic trajectory, implicitly allowing the model to account for vehicle momentum. As shown in Table 3, extending the context window yields considerable performance gains—most notably at the 5 s mark, where gear disambiguation and throttle prediction become significantly more reliable. Expanding further to 9 s provides only marginal improvements (Gear Macro F1 reaching 0.54), suggesting a 5 s acoustic window captures sufficient kinematic history to overcome transient ambiguities.

5. CONCLUSION

We presented F1Audio, a 300-hour dataset of Formula 1 engine audio features synchronized with car telemetry from 10 Grand Prix events of the 2024 season, alongside a complete cross-modal alignment pipeline. Our experiments with linear probes and MLP models establish a basic benchmark for these tasks. The temporal context ablation study demonstrates substantial performance improvements across all tasks, particularly throttle and gear.

We hope F1Audio provides a foundation for future research in automotive audio analysis, motorsport informatics, and cross-team engine acoustics.

6. ACKNOWLEDGMENT

The authors thank the Georgia Institute of Technology PACE GPU cluster team for computational resources.

⁵Source code: <https://github.com/GovindaMadhava/F1Audio>

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